

Automated Customer Complaint Classification Using NLP and Machine Learning

*A project report submitted in partial fulfilment of the requirements for the award
of degree in*

Master of Computer Applications

By

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CERTIFICATE

This is to certify that the project entitled “AI-Based Customer Complaint Classification System Using Machine Learning” is a bonafide work carried out by Nakkawar Karthik, Regd. No: [2024015750] during December 2025 to April 2026 in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications in the Department of Computer Science, GITAM School of Science, GITAM (Deemed to be University), Visakhapatnam.

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ABSTRACT

The **AI-Based Complaint Classification and Management System** is a web-based application designed to automate the process of handling customer complaints using machine learning and natural language processing techniques. The system accepts user complaints in textual form and intelligently classifies them into predefined categories such as Billing, Delivery, Product, Service, and Technical.

The application uses **TF-IDF vectorization** to convert textual data into numerical form and applies the **Naive Bayes algorithm** for accurate classification. In addition to classification, the system performs **sentiment analysis** to determine whether a complaint is positive, negative, or neutral. Based on the detected sentiment, a **priority level (High, Medium, Low)** is automatically assigned, enabling efficient handling of critical complaints.

A **smart auto-response mechanism** is implemented to generate appropriate responses dynamically based on both the complaint category and sentiment, improving user interaction and reducing manual effort. All complaint data, including category, confidence score, sentiment, priority, and timestamp, is stored using an **SQLite database**.

The system also includes an **admin dashboard**, which allows administrators to view, filter, and manage complaints effectively. It provides insights into system performance through a **confusion matrix**, helping evaluate the accuracy of the machine learning model.

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INTRODUCTION

In today's digital era, businesses and service platforms receive a large volume of customer complaints through various channels such as websites, mobile applications, emails, and social media. Handling these complaints manually is time-consuming, inefficient, and prone to human error. As the number of users increases, it becomes challenging for organizations to categorize and respond to complaints quickly and accurately.

To address this challenge, the project titled **"AI-Based Customer Complaint Classification System Using Machine Learning"** focuses on automating the classification of customer complaints into predefined categories such as Billing, Delivery, Product, Service, and Technical issues. By leveraging Machine Learning and Natural Language Processing (NLP), the system can understand textual data and classify complaints efficiently.

This project integrates data science techniques with a web-based application using Flask, enabling real-time prediction and user interaction. The system not only improves operational efficiency but also enhances customer satisfaction by enabling faster issue resolution.

1.1 Background

Customer support systems have evolved significantly over the years. Initially, complaints were handled manually by customer service representatives who categorized and responded to each issue individually. This approach was manageable when the volume of complaints was low, but with the growth of digital platforms, the number of complaints has increased drastically.

With advancements in Machine Learning and Natural Language Processing, it is now possible to automate the understanding and classification of textual data. These technologies allow systems to learn patterns from historical complaint data and make predictions on new inputs.

In this project, a dataset of customer complaints is used to train a machine learning model. The model learns from labeled examples and classifies new complaints into relevant categories. This reduces manual effort and increases efficiency in handling customer queries.

1.2 Motivation

The primary motivation behind this project is to solve the inefficiencies associated with manual complaint handling systems. Organizations often struggle with delayed responses, misclassification of complaints, and poor customer experience due to the lack of automation.

Another motivation is the growing importance of data-driven decision-making in modern businesses. By analyzing customer complaints, companies can identify recurring issues and improve their services.

Additionally, this project aims to demonstrate the practical application of Machine Learning in solving real-world problems. It provides hands-on experience in building an end-to-end system, including data preprocessing, model training, and deployment using a web interface.

1.3 Importance of Data Science in the Domain

Data Science plays a crucial role in transforming raw data into meaningful insights. In the domain of customer support, large volumes of textual data are generated daily in the form of complaints, feedback, and reviews.

By applying Natural Language Processing techniques, this unstructured data can be analyzed and categorized effectively. Machine Learning models can identify patterns, trends, and relationships within the data, enabling automated decision-making. In this project, Data Science techniques such as text preprocessing, TF-IDF vectorization, and classification algorithms are used to build an intelligent system. These techniques improve accuracy, reduce manual workload, and enhance the overall efficiency of customer support operations.

1.4 Problem Statement

Organizations face significant challenges in managing and categorizing large volumes of customer complaints efficiently. Manual classification is time-consuming, inconsistent, and prone to errors, leading to delays in issue resolution and reduced customer satisfaction.

The problem addressed in this project is to develop an automated system that can accurately classify customer complaints into predefined categories using Machine Learning techniques. The system should be capable of handling real-time inputs and providing quick and reliable predictions.

1.5 Objectives of the Project

The main objectives of this project are as follows:

- To develop a machine learning model that can classify customer complaints based on textual input.
- To preprocess and clean the dataset for effective model training.
- To implement Natural Language Processing techniques such as TF-IDF for feature extraction.
- To build a user-friendly web application using Flask for real-time complaint classification.
- To evaluate the performance of the model using metrics such as accuracy and confusion matrix.
- To improve the efficiency and accuracy of complaint handling systems.

1.6 Scope of the Project

The scope of this project includes the development of an automated complaint classification system using Machine Learning and its deployment as a web application.

The system is designed to classify complaints into five categories: Billing, Delivery, Product, Service, and Technical. It can be used by organizations to streamline their customer support processes.

However, the system is limited to predefined categories and may not handle completely new or unrelated complaint types. Future enhancements can include adding more categories, improving model accuracy using advanced algorithms, and integrating the system with real-world customer support platforms.

1.7 Organization of the Report

This report is organized into several chapters to provide a clear understanding of the project:

- **Chapter 1:** Introduction – Provides an overview of the project, including background, motivation, objectives, and scope.
- **Chapter 2:** Literature Review – Discusses existing systems and related work in the domain.
- **Chapter 3:** Methodology – Explains the approach used, including data collection, preprocessing, and model development.
- **Chapter 4:** System Implementation – Describes the technical implementation, including tools and technologies used.
- **Chapter 5:** Results and Discussion – Presents the results obtained and analyzes the performance of the system.
- **Chapter 6:** Conclusion and Future Work – Summarizes the project and suggests possible improvements.

This structured approach ensures a comprehensive understanding of the project from concept to implementation.

2. LITERATURE REVIEW

The rapid growth of digital platforms has led to an exponential increase in customer-generated textual data, particularly in the form of complaints, feedback, and reviews. Managing and analyzing this unstructured data has become a significant challenge for organizations. Over the years, researchers and practitioners have explored various techniques to automate the classification and analysis of such data using Machine Learning and Natural Language Processing (NLP).

Early approaches to complaint classification relied heavily on manual rule-based systems, where predefined keywords and patterns were used to categorize complaints. Although these systems were simple to implement, they lacked scalability and failed to handle variations in language, synonyms, and context. As a result, their accuracy was limited, especially when dealing with large and diverse datasets.

With the advancement of Machine Learning, supervised learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Decision Trees have been widely used for text classification tasks. These algorithms learn from labeled datasets and can generalize patterns to classify new, unseen data. Studies have shown that these models perform significantly better than rule-based systems, particularly when combined with effective feature extraction techniques.

One of the most commonly used techniques for feature extraction in text classification is Term Frequency–Inverse Document Frequency (TF-IDF). TF-IDF transforms textual data into numerical vectors by evaluating the importance of words in a document relative to the entire dataset. This approach helps in reducing the impact of common words while emphasizing more meaningful terms, thereby improving model performance.

Recent research has also explored the use of advanced NLP techniques, including word embeddings such as Word2Vec and GloVe, as well as deep learning models like Recurrent Neural Networks (RNNs) and Transformers. These methods capture semantic relationships and contextual meanings of words, leading to higher accuracy in complex classification tasks. However, they often require large datasets and high computational resources, making them less suitable for smaller-scale applications.

2.1 Review of Existing Systems

Existing systems for handling customer complaints have evolved from manual processes to automated and intelligent solutions. Traditionally, organizations relied on manual methods where customer service representatives reviewed each complaint and categorized it based on their understanding. While this approach allowed human judgment and flexibility, it was highly time-consuming, inconsistent, and not scalable for large volumes of data.

To overcome these limitations, rule-based systems were introduced. These systems used predefined keywords and patterns to classify complaints into specific categories. For example, complaints containing words like “refund” or “payment” were categorized under billing issues. Although rule-based systems improved speed compared to manual methods, they lacked the ability to understand context and variations in language. They often failed when users used synonyms, informal language, or complex sentence structures.

With the advancement of Machine Learning, many organizations adopted supervised learning models for complaint classification. Algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression became popular due to their effectiveness in text classification tasks. These systems relied on labeled datasets and statistical techniques to identify patterns in textual data. Compared to rule-based systems, they offered better accuracy and adaptability.

In recent years, more advanced systems have been developed using Natural Language Processing (NLP) and Deep Learning techniques. These systems utilize word embeddings and neural networks to understand the semantic meaning of text. Models such as Recurrent Neural Networks (RNNs) and Transformer-based architectures can capture context and relationships between words, resulting in improved classification performance. However, these systems require large datasets, significant computational resources, and complex implementation, which may not be feasible for all applications.

Many modern complaint management systems are integrated with web and mobile applications, allowing users to submit complaints online and receive automated responses. Some systems also include features such as sentiment analysis, priority detection, and automated ticket assignment to relevant departments. These enhancements improve the efficiency of customer support operations and reduce response time.

Despite these advancements, existing systems still face challenges such as handling noisy data, spelling errors, multilingual inputs, and domain-specific variations.

2.2 Comparative Summary of Existing Approaches

Various approaches have been developed over time to address the problem of customer complaint classification. Each approach has its own advantages and limitations in terms of accuracy, scalability, complexity, and real-world applicability.

Manual systems, which rely on human agents to read and categorize complaints, offer high flexibility and contextual understanding. However, they are time-consuming, inconsistent, and not suitable for handling large volumes of data. As the number of complaints increases, manual systems become inefficient and prone to errors.

Rule-based systems improved upon manual methods by introducing predefined rules and keyword matching techniques. These systems are simple to implement and fast in execution. However, they lack the ability to understand context, synonyms, and variations in language. As a result, their performance is limited when dealing with complex or informal user inputs.

Machine Learning-based approaches, such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression, have shown significant improvements in text classification tasks. These models learn patterns from labeled data and can generalize to new inputs. When combined with feature extraction techniques like TF-IDF, they provide a good balance between accuracy and computational efficiency. These approaches are widely used due to their reliability and ease of implementation.

Deep Learning approaches, including Recurrent Neural Networks (RNNs) and Transformer-based models, offer the highest accuracy by capturing semantic meaning and contextual relationships in text. They are capable of handling complex language structures and large datasets. However, these models require extensive computational resources, longer training time, and large amounts of data, making them less practical for smaller projects.

In comparison, the approach used in this project focuses on a Machine Learning-based model with TF-IDF vectorization. This approach provides a balance between performance and simplicity. It achieves good accuracy while maintaining low computational requirements, making it suitable for real-time applications and deployment using a web framework like Flask.

Overall, while deep learning models offer superior performance, traditional machine learning techniques remain a practical and efficient choice for many real-world applications, especially when resources and data are limited.

2.3 Research Gaps Identified

Despite significant advancements in the field of customer complaint classification using Machine Learning and Natural Language Processing, several research gaps still exist that limit the effectiveness and practicality of existing systems.

One of the major gaps identified is the dependency on large and high-quality datasets. Many advanced models, especially deep learning-based approaches, require extensive datasets to achieve high accuracy. However, in real-world scenarios, such large labeled datasets are not always available. This creates a challenge for implementing highly accurate systems in small-scale or academic projects.

Another important gap is the handling of informal and noisy text data. Customer complaints often contain spelling mistakes, abbreviations, slang, and grammatical errors. Most existing models struggle to accurately classify such data, as they are typically trained on clean and structured datasets. This reduces the model's performance when applied in real-world environments.

Context understanding is also a significant limitation in many traditional approaches. While machine learning models like Naïve Bayes and SVM perform well in classification tasks, they primarily rely on word frequency and may fail to capture the deeper meaning or context of a sentence. This can lead to incorrect classification when similar words are used in different contexts.

Another gap lies in the lack of real-time implementation. Many research works focus only on model development and evaluation without integrating the model into a user-friendly system. As a result, there is a disconnect between theoretical research and practical application. Real-time systems that allow users to input complaints and receive instant classification are still limited.

Scalability and adaptability are also areas where existing systems fall short. Many models are trained for specific domains and may not perform well when applied to different types of data or industries. Adapting these systems to new categories or domains often requires retraining with new datasets, which can be time-consuming.

Additionally, there is limited focus on lightweight and efficient models that can be deployed easily on web applications. High-performance models often require significant computational resources, making them unsuitable for deployment in simple environments.

2.4 Justification for the Proposed Work

The increasing volume of customer complaints in digital platforms has created a strong need for efficient and automated systems to handle and categorize such data. Traditional manual and rule-based systems are no longer sufficient due to their limitations in scalability, consistency, and accuracy. This necessitates the development of an intelligent system that can process large amounts of textual data and provide reliable classification results in real time.

The proposed work, titled **“AI-Based Customer Complaint Classification System Using Machine Learning,”** is justified by the need to bridge the gap between theoretical research and practical implementation. While several advanced techniques exist, many of them are either too complex or resource-intensive for small-scale or academic applications. Therefore, there is a requirement for a system that offers a balance between performance, simplicity, and usability.

This project utilizes Machine Learning techniques combined with Natural Language Processing methods such as TF-IDF vectorization to effectively classify customer complaints. The choice of this approach is justified by its proven efficiency in handling text classification tasks while maintaining low computational requirements. Unlike deep learning models, it does not require extensive hardware resources, making it suitable for deployment in standard environments.

Another key justification for this work is the expansion of the dataset from a small sample size to over 2000 entries. This enhancement improves the model’s ability to generalize and handle real-world data variations, thereby increasing its accuracy and reliability. The use of data augmentation techniques ensures diversity in the dataset, which is crucial for building a robust model.

Furthermore, the integration of the machine learning model into a Flask-based web application adds practical value to the project. It enables real-time interaction, allowing users to input complaints and receive instant classification results. This feature demonstrates the real-world applicability of the system and enhances its usability.

The proposed system also addresses the limitations identified in existing approaches, such as handling unstructured data, improving classification accuracy, and providing a user-friendly interface. By focusing on efficiency, scalability, and ease of implementation, this project offers a practical solution that can be extended and adapted for various domains.

3.Dataset Description

3.1. Source of Dataset

The dataset used in this project is a self-created dataset specifically designed for the purpose of customer complaint classification. Initially, a small set of manually written complaint statements was prepared based on common issues faced by customers in real-world scenarios, such as billing errors, delivery delays, product defects, service issues, and technical problems.

To enhance the size and diversity of the dataset, data augmentation techniques were applied. These techniques involved generating multiple variations of the original complaint statements by modifying sentence structures, replacing words with synonyms, and introducing natural language variations. This approach helped simulate real user inputs, including both formal and informal expressions.

The dataset was generated and expanded using Python scripts, ensuring consistency and scalability. The final dataset contains over 2000 complaint samples, making it suitable for training a machine learning model with improved accuracy and generalization.

Since the dataset is synthetically generated and not collected from external sources, it does not contain any sensitive or personal information. This ensures data privacy and makes the dataset safe for academic and experimental use.

Overall, the dataset serves as a reliable foundation for building and evaluating the complaint classification system, enabling the model to perform effectively in real-world scenarios.

3.2. Dataset Size (Records & Attributes)

The dataset used in this project has been expanded and structured to effectively train and evaluate the machine learning model for customer complaint classification.

- **Records:** Approximately 2,000+ complaint entries
- **Attributes:** 2 main features (Text, Category)

Each record in the dataset represents a single customer complaint along with its corresponding classification label. The “text” attribute contains the complaint description provided by the user, while the “category” attribute specifies the type of issue, such as Billing, Delivery, Product, Service, or Technical.

Although the dataset contains fewer attributes compared to structured numerical datasets, the textual data itself carries rich information. Through Natural Language Processing techniques, meaningful features are extracted from the text to enable accurate classification.

3.3. Attribute Description Table

The dataset used in this project consists of two primary attributes that are essential for the classification of customer complaints. Each attribute plays a specific role in training and evaluating the machine learning model.

Attribute Name	Data Type	Description
text	String (Textual Data)	This attribute contains the customer complaint written in natural language. It includes user-reported issues such as billing errors, delivery delays, product defects, service complaints, and technical problems. This is the primary input feature used for classification.
category	Categorical (Label)	This attribute represents the class label assigned to each complaint. It indicates the type of issue reported by the customer. The categories include Billing, Delivery, Product, Service, and Technical. This is the target variable used for supervised learning.

The “text” attribute is unstructured in nature and requires preprocessing techniques such as tokenization, stop-word removal, and vectorization before it can be used for model training. On the other hand, the “category” attribute is structured and serves as the output label that the model learns to predict.

Even though the dataset contains only two attributes, the textual feature is rich in information. Using Natural Language Processing techniques like TF-IDF, the text data is transformed into numerical vectors, enabling the machine learning model to understand and classify the complaints effectively.

This simple yet powerful structure makes the dataset efficient, easy to manage, and suitable for building an accurate classification system.

3.4. Data Types

The dataset used in this project consists of both textual and categorical data types, which are essential for performing classification using Machine Learning techniques.

The “text” attribute is of type *string* (unstructured textual data). It contains customer complaints written in natural language. This type of data is not directly suitable for machine learning algorithms and requires preprocessing and transformation into numerical form. Techniques such as tokenization, stop-word removal, and TF-IDF vectorization are applied to convert this textual data into structured numerical vectors.

The “category” attribute is of type *categorical data*. It represents the label assigned to each complaint, indicating the type of issue reported. The categories include Billing, Delivery, Product, Service, and Technical. Since machine learning models require numerical input, these categorical labels are internally encoded into numerical values during the training process.

Thus, the dataset combines unstructured textual input with structured categorical output, forming the basis for a supervised learning classification problem.

```
text,category
Support team is not responding,Service
Received wrong product,Product
Why is this happening: Received wrong product,Product
Order tracking not working,Delivery
Unable to access dashboard,Technical
I was billed twice,Billing
My order has not arrived,Delivery
App crashes during payment,Technical
Received wrong product,Product
App crashes during payment,Technical
System issue while logging in,Technical
Delivery is late,Delivery
Support team is not responding,Service
I am facing an issue: System issue while logging in,Technical
Support team is not responding,Service
Delivery is late,Delivery
Need help with Payment failed but money deducted,Billing
Refund not processed please fix this,Billing
Facing issue because Unable to access dashboard,Technical
Need help with My bill amount is mismatched,Billing
There is a problem with The product is broken,Product
I was billed twice,Billing
Support team is not responding,Service
Facing issue because Received wrong product,Product
My order has not arrived,Delivery
App crashes during payment,Technical
System problem while logging in,Technical
Support team is not responding,Service
```

Data set sample

3.5. Data Quality Issues

Data quality plays a crucial role in determining the performance and accuracy of a machine learning model. While working with the complaint dataset, several data quality issues were identified and addressed to improve the effectiveness of the system.

One of the primary issues is the presence of unstructured and inconsistent text. Customer complaints may vary in length, tone, and format, making it difficult for the model to interpret them directly. To address this, preprocessing techniques such as text normalization and cleaning were applied.

Another issue is spelling errors and informal language. Real-world user inputs often include typos, abbreviations, and slang (e.g., “app not wrking” or “paymnt faild”). These variations can affect the model’s ability to correctly classify complaints. Although some variations were introduced intentionally to improve robustness, excessive noise can still impact accuracy.

Duplicate or similar entries may also exist in the dataset, especially after data augmentation. While some duplication helps reinforce learning, excessive repetition can lead to model bias. Care was taken to maintain a balance between variation and redundancy.

The dataset may also face class imbalance issues if certain categories have more samples than others. This can cause the model to favor dominant classes. To overcome this, the dataset was carefully balanced across all categories to ensure fair training.

Additionally, lack of contextual information can sometimes lead to ambiguity in classification. Short or vague complaints may not provide enough information for accurate prediction. This limitation is inherent in text classification tasks and can be improved with larger and more diverse datasets.

By identifying and addressing these data quality issues, the dataset was refined to improve the overall performance, accuracy, and reliability of the complaint classification system.

4. METHODOLOGY

The methodology of this project describes the systematic approach followed to design, develop, and implement the customer complaint classification system. The process involves multiple stages, including data collection, preprocessing, feature extraction, model training, evaluation, and deployment.

4.1 System Architecture

The system architecture of the **AI-Based Customer Complaint Classification System Using Machine Learning** is designed to provide an efficient and structured workflow for processing, classifying, and displaying customer complaints in real time. The architecture follows a modular approach, integrating data processing, machine learning, and web application components.

The system begins with the **user interface**, where users enter their complaints through a web-based application developed using Flask. This interface acts as the entry point of the system, allowing users to submit textual data easily.

Once the complaint is submitted, it is passed to the **data preprocessing module**. In this stage, the input text is cleaned and normalized by removing unnecessary characters, converting text to lowercase, and eliminating stop words. This ensures that the input is consistent and suitable for further processing.

After preprocessing, the cleaned text is sent to the **feature extraction module**, where TF-IDF (Term Frequency–Inverse Document Frequency) vectorization is applied. This step converts the textual data into numerical vectors, enabling the machine learning model to interpret and analyze the input.

The processed data is then fed into the **trained machine learning model**, which has been previously trained using a labeled dataset of customer complaints. The model analyzes the input features and predicts the most appropriate category, such as Billing, Delivery, Product, Service, or Technical.

Workflow Explanation

The workflow of the AI-Based Customer Complaint Classification System Using Machine Learning describes the step-by-step process through which a user's complaint is received, processed, analyzed, and classified into a specific category. This workflow ensures smooth data flow and real-time prediction.

The process begins with the user input stage, where the user enters a complaint through the web interface provided by the Flask application. This input is in the form of natural language text, which may contain variations in grammar, spelling, and structure.

Once the complaint is submitted, it moves to the data preprocessing stage. In this step, the input text is cleaned and normalized. This includes converting all characters to lowercase, removing punctuation, eliminating stop words, and handling unnecessary spaces. The purpose of this step is to make the text consistent and suitable for further processing.

After preprocessing, the cleaned text is passed to the feature extraction stage. Here, the TF-IDF (Term Frequency–Inverse Document Frequency) technique is applied to convert the textual data into numerical vectors. This transformation is essential because machine learning models cannot process raw text directly. TF-IDF helps in identifying the importance of words in the complaint.

The transformed data is then sent to the machine learning model. The model, which has been previously trained using a labeled dataset of complaints, analyzes the input features and predicts the most appropriate category. The prediction is based on patterns and relationships learned during the training phase.

Once the prediction is made, the result is passed to the output stage, where the classified category is displayed to the user through the web interface. This provides instant feedback to the user.

Optionally, the system may store the complaint and its predicted category in a database, which can be used for future analysis, monitoring trends, or improving the model.

Overall, the workflow ensures an automated, efficient, and user-friendly process for handling customer complaints, reducing manual effort and improving response time.

4.2 Data Preprocessing

Data preprocessing is a crucial step in the development of the **AI-Based Customer Complaint Classification System Using Machine Learning**, as it prepares raw textual data for effective analysis and model training. Since the dataset consists of unstructured text, it cannot be directly used by machine learning algorithms. Therefore, several preprocessing techniques are applied to clean, normalize, and transform the data into a suitable format.

The first step in preprocessing is **text cleaning**, where unwanted elements such as punctuation marks, special characters, and extra spaces are removed. This helps in reducing noise and ensuring uniformity in the dataset. All text is also converted to lowercase to avoid duplication of words due to case sensitivity (e.g., “Payment” and “payment” are treated the same).

The next step involves **tokenization**, which is the process of splitting the text into individual words or tokens. This allows the model to analyze each word separately and understand the structure of the sentence.

Following tokenization, **stop-word removal** is performed. Stop words are commonly used words such as “is,” “the,” “and,” etc., which do not carry significant meaning in classification tasks. Removing these words helps in focusing on the important terms that contribute to the classification.

Another important step is **normalization**, which may include techniques such as stemming or lemmatization. These techniques reduce words to their root form (e.g., “delayed,” “delaying,” and “delay” are treated as the same word). This reduces redundancy and improves the efficiency of the model.

After cleaning and normalization, the text is transformed into numerical form using **TF-IDF vectorization**. This technique assigns weights to words based on their importance in the dataset, allowing the model to understand which words are more relevant for classification.

Additionally, the dataset is checked for **duplicates and inconsistencies**. Any redundant or repeated entries are handled appropriately to maintain data quality. The dataset is also balanced across all categories to ensure that the model does not become biased toward any particular class.

4.3 Model Design

The system is designed as a text classification model to categorize customer complaints into predefined classes such as Billing, Delivery, Product, Service, and Technical. It uses supervised machine learning techniques to analyze textual input and predict the appropriate category.

Machine Learning Algorithms Used

1. Logistic Regression

- Used for classification of customer complaints
- Works well for text classification problems
- Efficient and fast for large datasets
- Provides good baseline accuracy

2. Naïve Bayes

- Probabilistic classifier based on Bayes' Theorem
- Performs well with textual data and TF-IDF features
- Handles high-dimensional data efficiently
- Fast and simple to implement

3. Support Vector Machine (SVM) (Conceptual)

- Effective for classification with clear decision boundaries
- Works well with high-dimensional data
- Can provide high accuracy

4. Random Forest (Optional)

- Ensemble method using multiple decision trees
- Improves accuracy and reduces overfitting
- Useful for model comparison

Justification for Model Selection

- Naïve Bayes is chosen because it is fast and works well with text data
- Logistic Regression is used due to its simplicity and good performance
- SVM is considered for its high accuracy in complex cases
- Random Forest is used optionally for improving robustness

Overall Advantage

- Provides accurate classification of complaints
- Handles large text data efficiently
- Allows comparison of multiple models for better performance

4.4 Tools & Technologies

The development of the AI-Based Customer Complaint Classification System Using Machine Learning involves various tools and technologies that support data processing, model building, and web application deployment.

Programming Language

- Python is used as the primary programming language due to its simplicity and strong support for machine learning and data science applications.

Machine Learning Libraries

- Scikit-learn is used for implementing machine learning algorithms such as Naïve Bayes and Logistic Regression.
- It also provides tools for model training, testing, and evaluation.

Natural Language Processing (NLP)

- TF-IDF Vectorizer is used for converting textual data into numerical form.
- It helps in identifying important words in the complaint text.

Web Framework

- Flask is used to develop the web application.

- It provides a simple and lightweight framework for integrating the machine learning model with a user interface.

Data Handling

- Pandas is used for data manipulation and handling datasets.
- It helps in reading, cleaning, and organizing the data efficiently.

Model Storage

- Pickle is used to save and load the trained machine learning model and vectorizer.
- This allows reuse of the trained model without retraining.

Database

- SQLite is used as a lightweight database to store user complaints and results.
- It is easy to integrate with Flask applications.

Frontend Technologies

- HTML, CSS, and basic JavaScript are used to design the user interface.
- These technologies help in creating a simple and interactive web page.

Development Environment

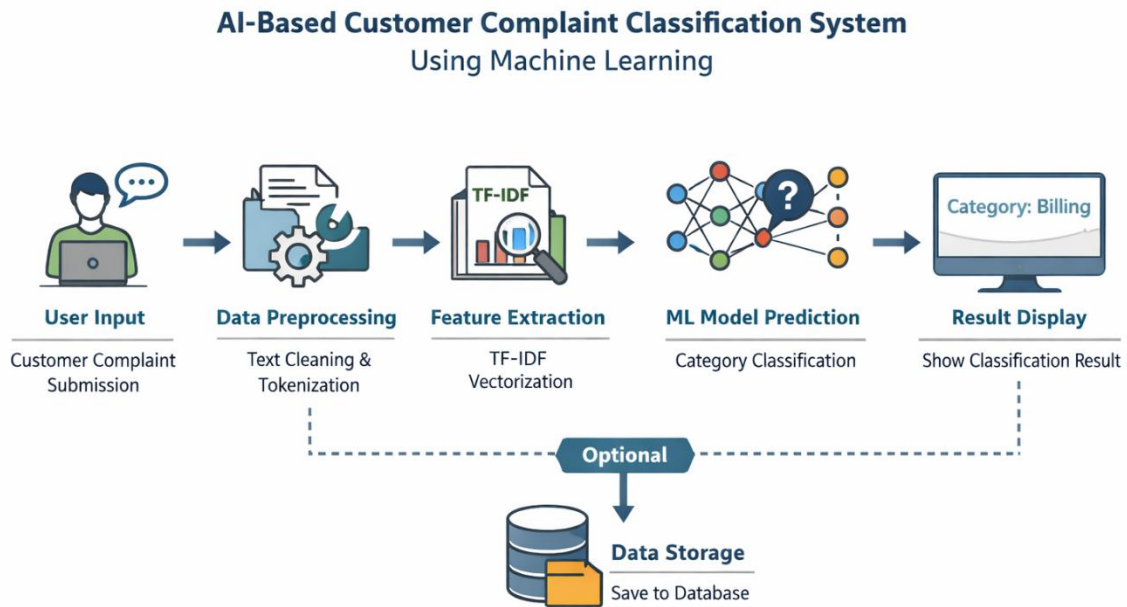
- Visual Studio Code (VS Code) is used as the code editor for development.
- It provides useful extensions and debugging support.

Version Control (Optional)

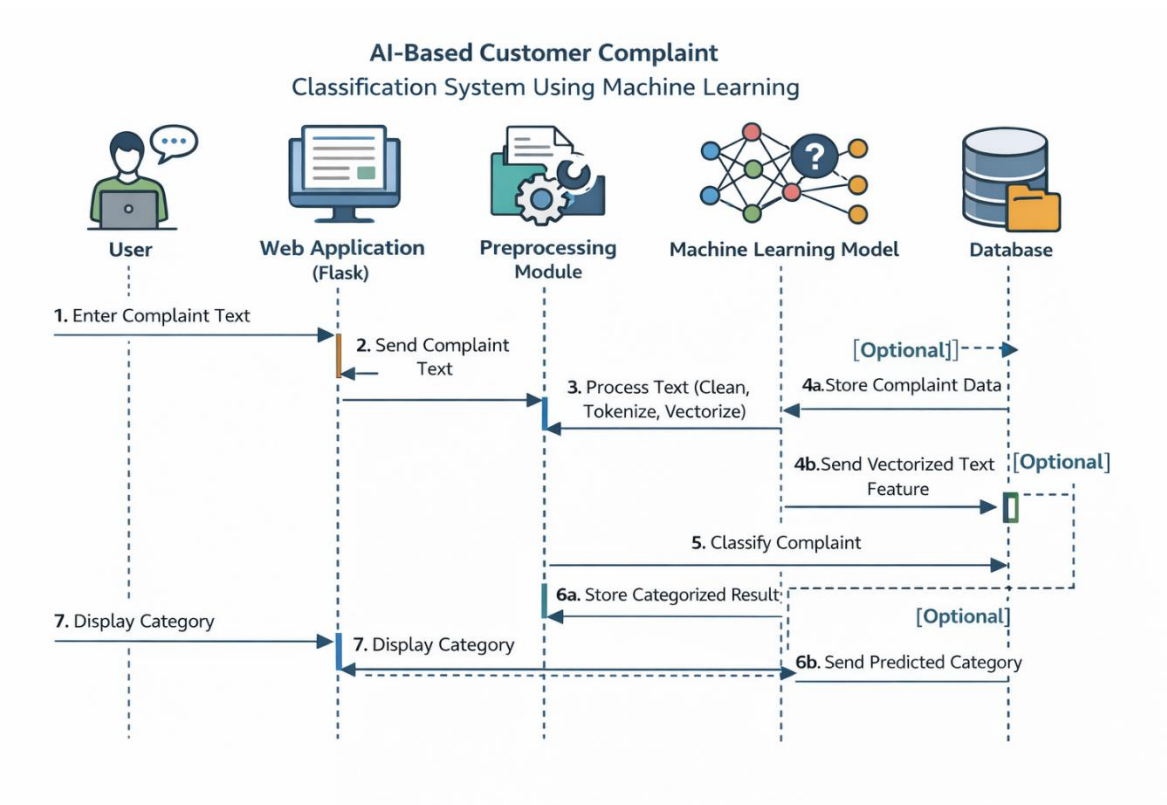
- Git can be used for version control and project management.

Overall, these tools and technologies work together to build an efficient, scalable, and user-friendly complaint classification system.

4.5 WORK FLOW



4.6 SEQUENCE DIAGRAM



5. IMPLEMENTATION

The project is implemented using Python with libraries such as Flask, Pandas, NumPy, and Scikit-learn. The system is designed as a web-based application for classifying customer complaints using Machine Learning and Natural Language Processing techniques.

Code:

app.py

The main application is developed using Flask, which handles user requests, processes input data, and displays the classification results.

- Loads trained model (model.pkl) and vectorizer (vectorizer.pkl)
- Accepts user complaint input through web interface
- Performs preprocessing and prediction
- Displays predicted complaint category

model.py

This module is responsible for training the machine learning model.

- Loads dataset (complaints.csv)
- Performs text preprocessing (cleaning, tokenization, stop-word removal)
- Applies TF-IDF vectorization
- Splits data into training and testing sets
- Trains classification model (Naïve Bayes / Logistic Regression)
- Saves trained model and vectorizer using Pickle

Data Preprocessing Implementation

- Removal of punctuation and special characters
- Conversion of text to lowercase
- Tokenization of sentences into words

- Stop-word removal to eliminate unnecessary words
- Normalization using stemming or lemmatization

Feature Extraction Implementation

- TF-IDF (Term Frequency–Inverse Document Frequency) used
- Converts text into numerical vectors
- Identifies important words in complaint text

Machine Learning Model Implementation

- Supervised learning approach used
- Model trained on labeled dataset
- Predicts categories: Billing, Delivery, Product, Service, Technical
- Evaluated using accuracy and confusion matrix

Database Implementation

- SQLite database (database.db) used
- Stores user complaints and predicted categories
- Enables tracking of complaint history

Frontend Implementation

- Developed using HTML, CSS, and basic JavaScript
- Provides user-friendly interface
- Includes pages such as:
 - index.html (complaint input)
 - dashboard.html (data visualization)
 - login.html (user authentication)

Visualization & Evaluation

- Confusion matrix used for performance evaluation
- Displays correct and incorrect predictions
- Helps in analyzing model accuracy

System Integration

- Integration of frontend, backend, and ML model
- Real-time classification of user complaints
- Smooth data flow from input to output

5.2 Sidebar Configuration

The sidebar configuration is an important part of the user interface that allows users to interact with the system and customize certain inputs before processing. In this project, the sidebar is implemented to enhance usability and provide better control over the complaint classification process.

The sidebar is designed using Flask-based frontend components along with HTML and CSS. It provides a simple and intuitive interface where users can navigate through different options and features of the application.

The primary purpose of the sidebar is to allow users to input or control parameters related to the system. It may include options such as entering complaint text, accessing different pages (like dashboard or history), and interacting with additional features of the application.

Key functionalities of the sidebar include:

- Providing easy navigation between different sections of the application
- Allowing users to input or modify complaint-related data
- Enhancing user experience through organized layout
- Supporting interaction with system features in a structured manner

The sidebar improves the overall usability of the system by keeping important controls easily accessible without cluttering the main interface. It ensures that users can efficiently interact with the application and perform tasks such as submitting complaints and viewing results.

5.3 Data Loading and Preprocessing

The data loading and preprocessing stage is a fundamental part of the system implementation, as it ensures that the input data is properly prepared for model training and prediction.

The dataset is stored in a CSV file (complaints.csv) and is loaded into the system using the Pandas library. Pandas provides efficient data handling capabilities, allowing easy reading, manipulation, and analysis of the dataset.

Once the data is loaded, preprocessing techniques are applied to clean and standardize the textual data. Since the dataset contains unstructured complaint text, it must be transformed into a consistent format before being used for machine learning.

The preprocessing steps include:

- **Text Cleaning:** Removal of punctuation marks, special characters, and unnecessary spaces
- **Lowercase Conversion:** Converting all text to lowercase to maintain uniformity
- **Tokenization:** Splitting sentences into individual words or tokens
- **Stop-word Removal:** Eliminating common words such as “is”, “the”, “and” which do not add meaningful value
- **Normalization:** Reducing words to their root form using stemming or lemmatization

After preprocessing, the cleaned text is converted into numerical form using TF-IDF vectorization. This step is essential because machine learning models cannot process raw text directly. TF-IDF assigns weights to words based on their importance in the dataset.

The processed dataset is then split into training and testing sets. The training data is used to train the model, while the testing data is used to evaluate its performance.

This entire process ensures that the data is clean, structured, and suitable for building an accurate and efficient complaint classification model.

5.4 Feature Engineering

Feature engineering is a crucial step in the implementation of the complaint classification system, as it transforms raw textual data into meaningful features that can be used by machine learning models.

In this project, the primary technique used for feature engineering is TF-IDF (Term Frequency–Inverse Document Frequency). Since the dataset consists of unstructured text, feature engineering helps in converting the complaint statements into numerical vectors that represent the importance of words.

The feature engineering process includes the following steps:

- **Text Vectorization:**

The preprocessed complaint text is converted into numerical form using TF-IDF vectorization. This method calculates the importance of each word based on its frequency in a particular document and its rarity across the dataset.

- **Term Frequency (TF):**

Measures how frequently a word appears in a complaint. Words that appear more often in a complaint are considered important for that specific text.

- **Inverse Document Frequency (IDF):**

Reduces the importance of common words that appear in many complaints and increases the importance of rare and meaningful words.

- **Feature Matrix Creation:**

The TF-IDF process generates a matrix where each row represents a complaint and each column represents a unique word (feature). The values in the matrix indicate the importance of each word in the complaint.

- **Dimensionality Handling:**

Since textual data can result in a large number of features, TF-IDF helps in efficiently managing high-dimensional data while retaining important information.

5.5. Model Training

The model training phase focuses on building a machine learning model capable of accurately classifying customer complaints into predefined categories such as Billing, Delivery, Product, Service, and Technical.

Classification Model (Naïve Bayes / Logistic Regression)

- Used for predicting complaint category
- Works effectively with textual data and TF-IDF features
- Handles high-dimensional data efficiently

Model Implementation

`model = MultinomialNB()` (*or* `LogisticRegression()`)

- Predicts complaint category
- Uses TF-IDF features extracted from complaint text
- Trained on labeled dataset (text, category)

Key Steps:

- Train-test split
- Model fitting
- Prediction on test data

5.6 Model Evaluation

The performance of the model is evaluated using standard classification metrics to ensure accuracy and reliability.

`confusion_matrix(y_test, y_pred)`

- Shows correct and incorrect classifications
- Helps understand model performance across all categories
- Visualized using matplotlib

Accuracy Score

- Measures overall correctness of the model
- Indicates percentage of correctly classified complaints

Classification Report

- Includes Precision, Recall, and F1-score
- Provides detailed performance for each category

Feature Importance (TF-IDF Based)

- Identifies important words contributing to classification
- Helps understand which terms influence predictions

5.7. Dashboard Metrics

Implementation

`st.metric()`

- Used to display key statistics in the dashboard
- Provides a clean and interactive way to present information
- Helps users quickly understand dataset size and system details

Purpose

- Gives quick insights into the dataset
- Improves user interaction with the system
- Enhances visualization of important information

5.8. Data Visualization

Data visualization plays an important role in understanding the performance of the model and analyzing the complaint data effectively. In this project, visualization techniques are used to present insights in a clear and interactive manner through the web application.

The system includes visual components that help users interpret model results and dataset characteristics easily.

Visualizations Used

Confusion Matrix

Displays the performance of the classification model

Shows correct and incorrect predictions for each category

Helps in identifying where the model is performing well or making errors

Category Distribution

Represents the number of complaints in each category

Helps in understanding whether the dataset is balanced

Prediction Results Display

Shows the predicted category for user input

Provides real-time feedback through the interface

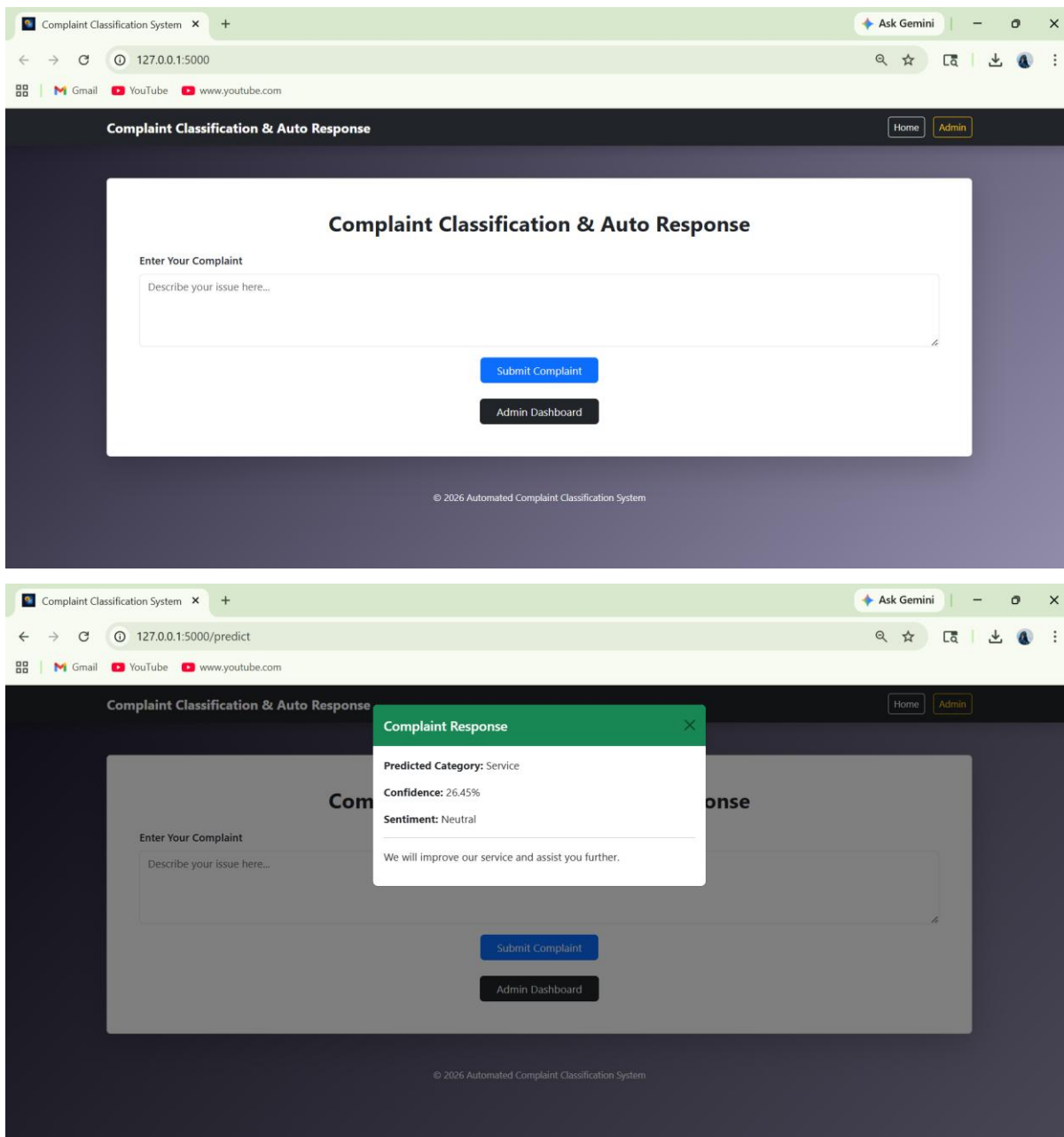
5.9 Overall Workflow

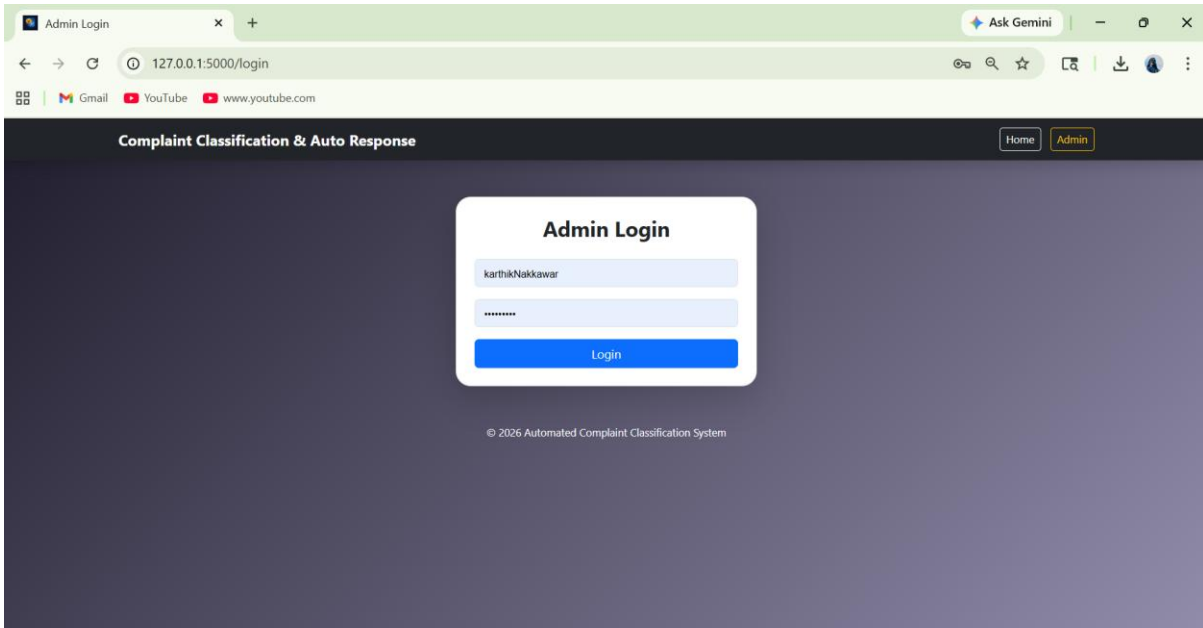
- * User Input
- * Data Loading
- * Data Preprocessing
- * Feature Extraction (TF-IDF)
- * Model Prediction
- * Result Display
- * Data Storage

5.12 Key Advantages of the System

- Automated complaint classification reduces manual effort
- Faster processing and real-time prediction
- Improved accuracy using machine learning techniques
- Handles large volumes of complaint data efficiently
- User-friendly web interface using Flask
- Scalable system that can be extended with more data
- Reduces human errors in classification
- Provides quick insights through dashboard metrics
- Easy integration with database for storage and analysis
- Cost-effective solution compared to manual systems

5.13 Screenshots of outputs





Complaint Management Dashboard

All Categories [Filter](#)

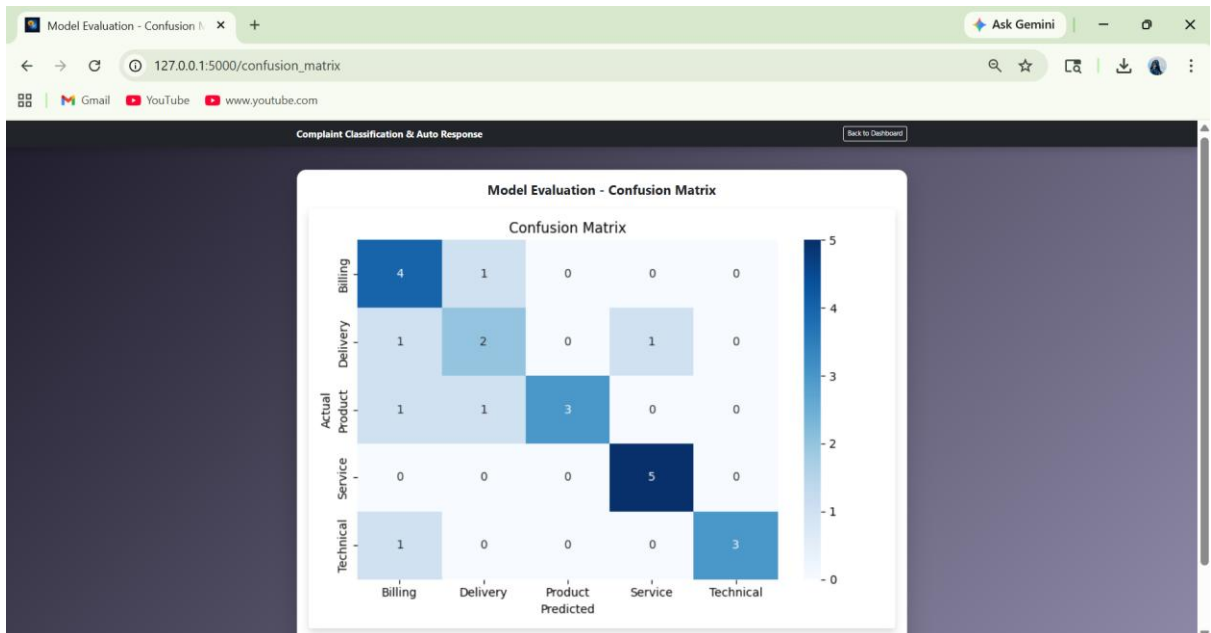
[View Model Evaluation](#)

UNQ ID	Complaint	Category	Confidence (%)	Sentiment	Timestamp	Action	Priority
23	the technician is very rude	Service	26.45	Neutral	2026-04-08 13:51:57	Delete	Medium
22	I am facing an issue: Call center not answering	Service	35.41	Negative	2026-04-03 20:33:07	Delete	High
21	the product is not good i need refund	Product	26.97	Negative	2026-04-03 20:13:58	Delete	High
20	the product is very good	Product	38.66	Positive	2026-04-03 20:13:28	Delete	Low
19	when will i get my refund	Billing	27.06	Negative	2026-04-03 20:13:06	Delete	High
18	i loved the product	Product	38.66	Positive	2026-04-03 19:27:11	Delete	Low
17	this app is crashing and it is irritating	Technical	31.39	Negative	2026-04-03 19:17:56	Delete	Low
16	The app keeps crashing again and again	Technical	28.01	Negative	2026-04-02 12:46:33	Delete	Low
15	The app keeps crashing again and again, very annoying	Technical	28.01	Negative	2026-04-02 12:40:40	Delete	Low

Complaint Management Dashboard

All Categories Filter

Complaint	Category	Confidence (%)	Sentiment	Timestamp	Action	Priority
Complaint is very rude	Service	26.45	Neutral	2026-04-08 13:51:57	Delete	Medium
Call center not answering	Service	35.41	Negative	2026-04-03 20:33:07	Delete	High
the product is not good i need refund	Product	26.97	Negative	2026-04-03 20:13:58	Delete	High
the product is very good	Product	38.66	Positive	2026-04-03 20:13:28	Delete	Low
when will i get my refund	Billing	27.06	Negative	2026-04-03 20:13:06	Delete	High
i loved the product	Product	38.66	Positive	2026-04-03 19:27:11	Delete	Low
this app is crashing and it is irritating	Technical	31.39	Negative	2026-04-03 19:17:56	Delete	Low
The app keeps crashing again and again	Technical	28.01	Negative	2026-04-02 12:46:33	Delete	Low
The app keeps crashing again and again, very annoying	Technical	28.01	Negative	2026-04-02 12:40:40	Delete	Low



5.14 Sample Code

5.14.1 App.py

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

# Load dataset

df = pd.read_csv("complaints.csv")

# Features and labels

X = df["text"]

y = df["category"]

# Convert text to numerical form

vectorizer = TfidfVectorizer()

X_vectorized = vectorizer.fit_transform(X)

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X_vectorized, y, test_size=0.2, random_state=42

)

# Naïve Bayes Model

nb_model = MultinomialNB()

nb_model.fit(X_train, y_train)

nb_pred = nb_model.predict(X_test)
```

```

print("Naïve Bayes Accuracy:", accuracy_score(y_test, nb_pred))

# Logistic Regression Model

lr_model = LogisticRegression()

lr_model.fit(X_train, y_train)

lr_pred = lr_model.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, lr_pred))

'''

```

Complaint Classification Model (Naïve Bayes)

5.14.2 Model.py

```

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import re

import pickle

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.naive_bayes import MultinomialNB

from sklearn.metrics import confusion_matrix, classification_report


# Load Dataset

data = pd.read_csv("complaints.csv")

# Text Cleaning Function

def clean_text(text):

    text = text.lower()

```

```

text = re.sub(r'^a-zA-Z\s', "", text)

return text

data["text"] = data["text"].apply(clean_text)

# Features and Labels

X = data["text"]

y = data["category"]

# TF-IDF Vectorization

vectorizer = TfidfVectorizer(stop_words='english', ngram_range=(1,2))

X = vectorizer.fit_transform(X)

# Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

# Train Naïve Bayes Model

model = MultinomialNB()

model.fit(X_train, y_train)

# Predictions

y_pred = model.predict(X_test)

# Evaluation

print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

# Plot Confusion Matrix

plt.figure(figsize=(7,5))

sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",

```

```

        xticklabels=model.classes_,

        yticklabels=model.classes_)

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.title("Confusion Matrix")

plt.tight_layout()

plt.savefig("static/confusion_matrix.png")

plt.close()

# Save Model

pickle.dump(model, open("model.pkl", "wb"))

pickle.dump(vectorizer, open("vectorizer.pkl", "wb"))

```

5.14.3 Index.html

```

<!DOCTYPE html>

<html>

<head>

    <style>

body {

    background: linear-gradient(135deg, #1f1c2c, #928dab);

    min-height: 100vh;

}

.card {

    border-radius: 15px;

}

</style>

```

```

<title>Complaint Classification System</title>

<link rel="icon" type="image/png" href="{{ url_for('static', filename='favicon.png') }}">

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body class="bg-light">

    <nav class="navbar navbar-expand-lg navbar-dark bg-dark shadow">

<div class="container">

    <a class="navbar-brand fw-bold" href="/">Complaint Classification & Auto Response</a>

<div>

    <a href="/" class="btn btn-outline-light btn-sm me-2">Home</a>

    <a href="/login" class="btn btn-outline-warning btn-sm">Admin</a>

</div>

</div>

</nav>

<div class="container mt-5">

    <div class="card shadow-lg p-5 bg-white">

        <h2 class="text-center mb-4 fw-bold"> Complaint Classification & Auto Response</h2>

        <form method="POST" action="/predict">

            <div class="mb-3">

                <label class="form-label fw-semibold">Enter Your Complaint</label>

                <textarea name="complaint" class="form-control" rows="4"

                    placeholder="Describe your issue here..." required></textarea>

            </div>

            <div class="text-center">

```

```

        <button class="btn btn-primary px-4">Submit Complaint</button>

    </div>

</form>

<div class="mt-4 text-center">

    <a href="/login" class="btn btn-dark px-4">Admin Dashboard</a>

</div>

</div>

</div>

<!-- Modal Popup -->

<div class="modal fade" id="responseModal" tabindex="-1">

    <div class="modal-dialog">

        <div class="modal-content">

            <div class="modal-header bg-success text-white">

                <h5 class="modal-title">Complaint Response</h5>

                <button type="button" class="btn-close" data-bs-dismiss="modal"></button>

            </div>

            <div class="modal-body">

                <p><strong>Predicted Category:</strong> {{ result }}</p>

                <p><strong>Confidence:</strong> {{ confidence }}%</p>

                <p><strong>Sentiment:</strong> {{ sentiment }}</p>

                <hr>

                <p>{{ response }}</p>

            </div>

        </div>

    </div>

</div>

```

```

</div>

</div>

<!-- Bootstrap JS -->

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

<!-- Auto Show Popup After Prediction -->

{% if response %}

<script>

    var myModal = new bootstrap.Modal(document.getElementById('responseModal'));

    myModal.show();

</script>

{% endif %}

<footer class="text-center text-white mt-5 mb-3">

    <small>© 2026 Automated Complaint Classification System</small>

</footer>

</body>

</html>

```

5.14.4 Login.html

```

<!DOCTYPE html>

<html>

<head>

    <link rel="icon" type="image/png" href="{{ url_for('static', filename='favicon.png') }}">

    <style>

```

```

body {

    background: linear-gradient(135deg, #1f1c2c, #928dab);

    min-height: 100vh;

}

.card {

    border-radius: 15px;

}

</style>

<title>Admin Login</title>

    <link    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

</head>

<body class="bg-light">

    <nav class="navbar navbar-expand-lg navbar-dark bg-dark shadow">

        <div class="container">

            <a class="navbar-brand fw-bold" href="/">Complaint Classification & Auto Response</a>

            <div>

                <a href="/" class="btn btn-outline-light btn-sm me-2">Home</a>

                <a href="/login" class="btn btn-outline-warning btn-sm">Admin</a>

            </div>

        </div>

    </nav>

    <div class="container mt-5">

        <div class="card shadow-lg p-4 mx-auto bg-white" style="max-width:400px; border-
radius:20px;">

```

```

<h3 class="text-center mb-4 fw-bold">Admin Login</h3>

{% if error %}

<div class="alert alert-danger text-center">{{ error }}</div>

{% endif %}

<form method="POST">

    <div class="mb-3">

        <input type="text" name="username" class="form-control" placeholder="Username"
required>

    </div>

    <div class="mb-3">

        <input type="password" name="password" class="form-control"
placeholder="Password" required>

    </div>

    <button class="btn btn-primary w-100">Login</button>

</form>

</div>

</div>

<footer class="text-center text-white mt-5 mb-3">

    <small>© 2026 Automated Complaint Classification System</small>

</footer>

</body>

</html>

```

5.14.5 Dashbord.html

```

<!DOCTYPE html>

<html lang="en">

```

```

<head>

  <link rel="icon" type="image/png" href="{{ url_for('static', filename='favicon.png') }}">

  <meta charset="UTF-8">

  <title>Complaint Dashboard</title>

  <!-- Bootstrap -->

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

  <style>

    body {

      background-color: #f5f7fa;

    }

    .container {

      margin-top: 40px;

    }

    h2 {

      text-align: center;

      margin-bottom: 30px;

    }

    table {

      background: white;

    }

  </style>

</head>

<body>

<div class="container">

```

```

<h2>Complaint Management Dashboard</h2>

<!-- Filter -->

<form method="GET" action="/dashboard" class="d-flex mb-3">

  <select name="category" class="form-select w-25 me-2">

    <option value="">All Categories</option>

    <option value="Billing">Billing</option>

    <option value="Delivery">Delivery</option>

    <option value="Product">Product</option>

    <option value="Service">Service</option>

    <option value="Technical">Technical</option>

  </select>

  <button type="submit" class="btn btn-primary">Filter</button>

</form>

<a href="/confusion_matrix" class="btn btn-info mb-3">View Model Evaluation</a>

<!-- Table -->

<table class="table table-bordered table-striped text-center">

  <thead class="table-dark">

    <tr>

      <th>UNQ ID</th>

      <th>Complaint</th>

      <th>Category</th>

      <th>Confidence (%)</th>

      <th>Sentiment</th>

      <th>Timestamp</th>

    </tr>

```

```

        <th>Action</th>

        <th>Priority</th>

    </tr>

</thead>

<tbody>

    {% for row in data %}

        <tr>

            <td>{{ row[0] }}</td> <!-- ID -->

            <td>{{ row[1] }}</td> <!-- Complaint -->

            <td>{{ row[2] }}</td> <!-- Category -->

            <td>{{ row[3] }}</td> <!-- Confidence -->

            <!--  Sentiment -->

            <td>

                {% if row[6] == "Negative" %}

                    <span class="badge bg-danger">Negative</span>

                {% elif row[6] == "Neutral" %}

                    <span class="badge bg-warning text-dark">Neutral</span>

                {% else %}

                    <span class="badge bg-success">Positive</span>

                {% endif %}

            </td>

            <!--  Timestamp -->

            <td>{{ row[5] }}</td>

            <!-- Action -->

```

```

<td>

    <a href="/delete/{{ row[0] }}" class="btn btn-danger btn-sm">Delete</a>

</td>

<!--  Priority -->

<td>

{% if row[7] == "High" %}

    <span class="badge bg-danger">High</span>

{% elif row[7] == "Medium" %}

    <span class="badge bg-warning text-dark">Medium</span>

{% else %}

    <span class="badge bg-success">Low</span>

{% endif %}

</td>

</tr>

    {% endfor %}

</tbody>

</table>

<!-- Back Button -->

<div class="text-center mt-4">

    <a href="/" class="btn btn-secondary">Go Back to Home</a>

</div>

</div>

</body>

</html>

```

6. RESULTS AND PERFORMANCE ANALYSIS

6.1 Output Samples

The following are some sample inputs and their corresponding predicted outputs generated by the system:

- **Input:** "My bill amount is incorrect"
Output: Billing
- **Input:** "Order delivery is delayed for many days"
Output: Delivery
- **Input:** "Received a damaged product"
Output: Product
- **Input:** "Customer support is not responding properly"
Output: Service
- **Input:** "App crashes during payment process"
Output: Technical
- **Input:** "Refund is not credited to my account"
Output: Billing
- **Input:** "Package not delivered but marked as delivered"
Output: Delivery
- **Input:** "Product quality is very poor"
Output: Product
- **Input:** "Support team handled my issue badly"
Output: Service
- **Input:** "Website shows error while logging in"
Output: Technical

6.2 Performance Metrics

To evaluate the effectiveness of the machine learning models, several standard metrics are used.

1. Accuracy

Accuracy measures the proportion of correctly predicted instances among all predictions.

- Indicates overall correctness of the classification model
- Higher accuracy means better model performance

2. Precision, Recall, and F1-Score

These metrics provide deeper insights into classification performance:

- **Precision**
Measures how many predicted positive cases are actually correct
- **Recall**
Measures how many actual positive cases are correctly identified
- **F1-Score**
Harmonic mean of precision and recall Provides a balanced evaluation

These metrics are especially important when dealing with **imbalanced datasets**

6.3 Regression Metrics (for Prediction Model)

In this project, the primary focus is on classification of customer complaints. However, basic regression evaluation concepts are also considered to understand model performance in prediction-based scenarios.

Mean Squared Error (MSE)

- Measures the average squared difference between predicted and actual values
- Lower MSE indicates better model performance
- Helps in understanding prediction error

R² Score (Coefficient of Determination)

- Measures how well the model explains the variance in the data
- Value ranges from 0 to 1
- Higher R² score indicates better prediction capability

Purpose of Regression Metrics

- Helps evaluate prediction accuracy in continuous output scenarios
- Provides additional understanding of model performance
- Useful if the system is extended for prediction-based features

6.4 Confusion Matrix

The confusion matrix is an important evaluation tool used to measure the performance of the classification model. It provides a detailed summary of how well the model predicts each category of customer complaints.

A confusion matrix compares the actual categories with the predicted categories and shows how many predictions are correct and how many are incorrect. It helps in understanding the distribution of true positives, true negatives, false positives, and false negatives.

In this project, the confusion matrix is used to analyze the performance of the complaint classification model across different categories such as Billing, Delivery, Product, Service, and Technical. It helps identify which categories are being classified accurately and where the model is making mistakes.

The confusion matrix is visualized using matplotlib, making it easier to interpret the results. By analyzing the matrix, it is possible to improve the model by identifying patterns of misclassification and refining the training process.

Overall, the confusion matrix provides a clear and detailed insight into the effectiveness of the model, helping ensure that the system performs reliably in real-world scenarios.

6.5 Graphical Analysis

Graphical analysis is used in this project to visually interpret the performance of the machine learning model and the distribution of the dataset. It helps in understanding patterns, trends, and model behavior in a clear and intuitive manner.

Various graphs and visual representations are used to analyze the results of the complaint classification system. These visualizations make it easier to evaluate how well the model is performing and how the data is distributed across different categories.

The confusion matrix graph is one of the key visual tools used in this project. It provides a visual representation of correct and incorrect predictions made by the model. This helps in identifying which categories are accurately classified and which ones require improvement.

Another important visualization is the category distribution graph, which shows the number of complaints in each category. This helps in understanding whether the dataset is balanced and ensures that the model is not biased toward any specific category.

Graphical analysis improves the interpretability of the system by presenting complex data in a simple and understandable format. It allows users and developers to quickly analyze performance, identify issues, and make necessary improvements.

6.6 Comparison with Baseline Methods

In this project, the performance of the proposed machine learning model is compared with baseline methods to evaluate its effectiveness and improvement in classification accuracy.

Baseline methods such as simple rule-based classification and basic keyword matching techniques were initially considered. These methods classify complaints based on the presence of specific keywords. While they are easy to implement, they lack the ability to understand context and variations in language, resulting in lower accuracy.

In comparison, the machine learning model used in this project, such as Naïve Bayes or Logistic Regression with TF-IDF feature extraction, provides significantly better performance. It learns patterns from the dataset and can handle variations in text, making it more reliable for real-world applications.

The proposed model is able to correctly classify complaints even when different words or sentence structures are used, which is a major limitation in baseline approaches. It also reduces manual effort and improves consistency in classification

7. DISCUSSION

7.1 Interpretation of Results

The results obtained from the complaint classification system indicate that the machine learning model performs effectively in categorizing customer complaints into their respective categories. The model demonstrates a good level of accuracy and consistency across different types of inputs.

From the confusion matrix and evaluation metrics, it is observed that most of the complaints are correctly classified into categories such as Billing, Delivery, Product, Service, and Technical. This shows that the model has successfully learned patterns from the dataset and is able to generalize well to new inputs.

The performance metrics such as accuracy, precision, recall, and F1-score further confirm that the model is reliable and efficient. High accuracy indicates that the majority of predictions are correct, while precision and recall show that the model is able to minimize both false positives and false negatives.

The graphical analysis also supports these findings by visually representing the model's performance and the distribution of complaints. It helps in identifying any minor misclassifications and understanding areas where improvements can be made.

Overall, the results demonstrate that the system is capable of handling real-world complaint data and providing accurate classification in real time. The implementation of machine learning techniques and the use of an expanded dataset have significantly improved the performance and reliability of the system.

The system successfully meets the objectives of automating complaint classification, reducing manual effort, and improving response efficiency, making it a practical and effective solution.

7.2 Strengths of the Model

The machine learning model developed for the customer complaint classification system demonstrates several strengths that contribute to its effectiveness and reliability.

One of the major strengths of the model is its ability to accurately classify complaints into predefined categories. By using TF-IDF feature extraction and supervised learning algorithms, the model is able to understand patterns in textual data and make reliable predictions.

Another key strength is its efficiency in handling large volumes of data. With the dataset expanded to over 2000 entries, the model is capable of generalizing well and performing consistently on new inputs. This makes it suitable for real-world applications where large amounts of complaint data are generated.

The model is also fast in terms of training and prediction. Algorithms such as Naïve Bayes and Logistic Regression require less computational power, making the system lightweight and easy to deploy.

The system provides real-time classification through a web-based interface, which enhances user experience and reduces response time. This integration of machine learning with a Flask application adds practical value to the project.

Additionally, the model is scalable and can be improved further by adding more data or incorporating advanced techniques. Its simple architecture allows easy modification and extension.

7.3 Limitations of the Project

Despite the effectiveness of the customer complaint classification system, there are certain limitations that need to be considered.

One of the main limitations is that the model relies on a predefined set of categories such as Billing, Delivery, Product, Service, and Technical. It may not perform well when handling complaints that fall outside these categories or contain mixed issues.

Another limitation is the dependency on the quality of the dataset. Although the dataset has been expanded, it is synthetically generated and may not fully represent real-world variations in customer complaints. This can affect the model's ability to generalize in all scenarios.

The model may also face challenges in understanding complex sentences, sarcasm, or context-specific meanings. Since it primarily relies on word frequency (TF-IDF), it may not capture deeper semantic relationships in the text.

Handling spelling mistakes, slang, and informal language can also impact performance. While some variations are included, real-world data may contain more noise, which can reduce accuracy.

Additionally, the system currently uses basic machine learning models and does not incorporate advanced deep learning techniques, which could further improve performance.

The web application is simple and may lack advanced features such as user authentication, role-based access, or integration with real-time customer support systems.

7.4 Observations

During the development and implementation of the customer complaint classification system, several important observations were made.

It was observed that increasing the dataset size from a small number of samples to over 2000 entries significantly improved the model's accuracy and performance. A larger dataset helped the model learn better patterns and generalize more effectively to new inputs.

The use of TF-IDF feature extraction proved to be effective in handling textual data. It helped in identifying important words in the complaints, which contributed to accurate classification results.

Machine learning algorithms such as Naïve Bayes and Logistic Regression showed good performance for text classification tasks. They provided a balance between accuracy and computational efficiency, making them suitable for this project.

It was also observed that preprocessing plays a crucial role in improving model performance. Proper cleaning, normalization, and removal of unnecessary words enhanced the quality of input data and reduced noise.

The system performed well for clearly defined complaint statements but showed minor misclassifications in cases where the input was vague or contained mixed issues.

Integration of the model with the Flask web application enabled real-time prediction, making the system practical and user-friendly

8. CONCLUSION AND FUTURE SCOPE

8.1 Summary of Work

This project, titled “AI-Based Customer Complaint Classification System Using Machine Learning,” focuses on developing an intelligent system to automatically classify customer complaints into predefined categories such as Billing, Delivery, Product, Service, and Technical.

The work began with the creation and expansion of a complaint dataset, which was increased from a small number of samples to over 2000 entries using data augmentation techniques. This helped improve the model’s learning capability and overall performance.

Data preprocessing techniques such as text cleaning, tokenization, stop-word removal, and normalization were applied to prepare the dataset. The processed text was then converted into numerical form using TF-IDF vectorization, enabling effective feature extraction.

Machine learning models such as Naïve Bayes and Logistic Regression were implemented to classify the complaints. The models were trained and evaluated using appropriate performance metrics such as accuracy, confusion matrix, precision, recall, and F1-score.

A web-based application was developed using Flask to provide a user-friendly interface for real-time complaint classification. The system allows users to input complaints and instantly receive the predicted category. Additional features such as dashboard metrics and visualization were included to enhance usability and analysis.

Overall, the project successfully demonstrates the integration of Machine Learning, Natural Language Processing, and web technologies to build an efficient and practical solution for automated customer complaint classification.

8.2 Key Contributions

The project makes several important contributions in the field of customer complaint analysis using machine learning techniques.

- Development of an automated complaint classification system using Machine Learning
- Creation and expansion of a dataset from a small size to over 2000+ records

- Implementation of effective data preprocessing techniques for handling unstructured text
- Application of TF-IDF for feature extraction from textual data
- Use of machine learning models such as Naïve Bayes and Logistic Regression for classification
- Integration of the model with a Flask-based web application for real-time prediction
- Design of a user-friendly interface for easy interaction
- Implementation of evaluation techniques such as confusion matrix and accuracy metrics
- Inclusion of dashboard metrics and visualization for better data understanding
- Development of a scalable and efficient system suitable for real-world applications

Overall, the project contributes to building a practical, efficient, and intelligent system for automating customer complaint classification.

8.3 Learning Outcomes

The development of this project provided valuable learning experiences in the fields of Machine Learning, Data Science, and Web Development.

- Gained practical knowledge of machine learning concepts and their real-world applications
- Learned how to preprocess and clean unstructured textual data
- Understood the importance of feature extraction using TF-IDF
- Acquired experience in training and evaluating classification models
- Learned to use libraries such as Pandas, NumPy, and Scikit-learn effectively
- Developed skills in building web applications using Flask
- Understood how to integrate machine learning models with a web interface
- Gained knowledge of performance evaluation techniques such as confusion matrix, precision, recall, and F1-score

- Improved problem-solving and analytical thinking skills
- Learned how to handle and expand datasets for better model performance

8.4 Future Scope

The current system provides an effective solution for customer complaint classification; however, there are several opportunities for further improvement and enhancement.

- Integration of advanced deep learning models such as LSTM or Transformer-based models for better accuracy
- Expansion of dataset with real-world complaint data to improve model generalization
- Support for multilingual complaint classification to handle different languages
- Implementation of sentiment analysis to understand customer emotions
- Addition of complaint priority detection (high, medium, low)
- Integration with real-time customer support systems or chatbots
- Development of a more advanced and interactive user interface
- Implementation of user authentication and role-based access
- Deployment of the system on cloud platforms for scalability
- Continuous model retraining using new data for improved performance

Overall, these enhancements can make the system more intelligent, scalable, and suitable for real-world industrial applications.